

Error Correction – Hamming code

Parity bit - One bit error

CRC – Cyclic redundancy check – Burst error

Ex – 10001 – Count of the bit – even – Even parity

10011 – odd parity bit

10100 – Data – Sender side

101000 – Even parity

Received data – 111000

Retransmission – way to correct the error

Odd parity - 10101 - 101011 –

Receiver - 100111

Hamming code – Detect and correct the single bit error

Data (m) – Any length - 1011

Redundant bit r - Length of r or Parity bit

$2^r \geq m+r+1$ - Place the redundant bit at the power of 2

$$p1 = 2^0$$

$$p2 = 2^1$$

$$p3 = 2^2$$

$$p4 = 2^4$$

R1 R2 R3 R4

2^0 2^1 2^2 2^4

Example

Data $m = 1001$ = length of $m = 4$

$$2^r \geq m+r+1$$

$$= 4+1+1 = 6$$

$2=6 \rightarrow$ wrong

$$2^2 = 4+2+1$$

$4 = 7 \rightarrow$ Wrong

$$2^3 = 4+3+1$$

$$8 = 8$$

Length of the parity bit = 3 - P8 P4 P2 P1

Length of the original data = $4+3 = 7$

1001

7	6	5	4	3	2	1
1	0	0	P4	1	P2	P1

Binary value

	P4	P2	P1	
1	0	0	1	P1 – 1, 3, 5 7
2	0	1	0	P2 - 2,3,6,7
3	0	1	1	P4 – 4,5,6,7

4 1 0 0

5 1 0 1

6 1 1 0

7 1 1 1

7	6	5	4	3	2	1
1	0	0	P4	1	P2	P1

P1 – 1 , 3, 5 7

P1 = p1 1 0 1 - **0** 1 1 0 – Even parity

P2 - 2,3,6,7

P2 – p2 1 0 1 - **0** 1 0 1 – Even parity

P4 – 4 5 6 7

P4 - p4 0 0 1 - **1** 0 0 1

Consider the even parity bit transmission from sender to receiver

p4 p2 p1

1 0 0

7	6	5	4	3	2	1
1	0	0	1	1	0	0

	7	6	5	4	3	2	1
Sender -	1	0	0	1	1	0	0
Receiver -	1	1	0	1	1	0	0
P1 - 1, 3, 5, 7	-	0	1	0	1	-	0

P2 - 2, 3, 6, 7 - 0 1 1 1 - 1

P4 - 4, 5, 6, 7 - 1 0 1 1 - 1

p4 p2 p1

1 1 0 - 6